Comprehensive Campus Environmental and Carbon Audit Report 2024

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COMPREHENSIVE CAMPUS ENVIRONMENTAL AND CARBON AUDIT REPORT 2024

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Conducting Authority: Bureau of Energy and Carbon Verification in coordination with VESIT Campus Sustainability Taskforce.

Audited Entity: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Chembur, Mumbai.

1. EXECUTIVE SUMMARY AND AUDIT METHODOLOGY

An exhaustive calendar year 2024 baseline greenhouse gas (GHG) and utility audit was conducted across the 32,500 m2 built-up campus of Vivekananda Education Society's Institute of Technology (VESIT Green Campus) in Chembur, Mumbai. The accounting methodology adheres rigorously to the GHG Protocol Corporate Accounting and Reporting Standard and ISO 14064-1 specifications. The audit quantified Scope 1 direct emissions, Scope 2 indirect emissions from imported electricity, and material Scope 3 value-chain emissions generated across seven principal facilities: Academic Block A (BLD_ACA), Academic Block B (BLD_ACB), Computer Laboratories (BLD_LAB), Library (BLD_LIB), Auditorium (BLD_AUD), Canteen (BLD_CAN), and Administration Block (BLD_ADM).

The verified institutional carbon footprint for calendar year 2024 stands at exactly 2,450,000 kg CO2e (2,450.00 metric tonnes CO2e), establishing the baseline against which the campus net-zero target (TRG_NET_ZERO) for December 31, 2040, is statutorily benchmarked.

2. EMISSION SOURCE ATTRIBUTION AND ACTIVITY INVENTORY

2.1 Scope 1 (Direct Combustion Emissions):

Direct emissions are predominantly generated by stationary diesel generator sets (DG) deployed for emergency electrical backup during regional grid outages. In 2024, emergency generators consumed 18,400 litres of low-sulfur high-speed diesel fuel. Applying documented IPCC Emission Factor EF_DIESEL_GEN (2.68 kg CO2e/litre), Scope 1 emissions totaled 49,312 kg CO2e (2.01% of gross institutional emissions). An additional 1,200 kg CO2e was generated from minor liquefied petroleum gas (LPG) usage in academic laboratories.

2.2 Scope 2 (Indirect Emissions from Purchased Grid Electricity):

Campus electricity importation from the Maharashtra State regional grid totaled 2,050,000 kWh in 2024. In accordance with Central Electricity Authority (CEA) Baseline Database Version 19.0 (Emission Factor EF_ELEC_GRID = 0.82 kg CO2e/kWh), imported electricity accounted for 1,681,000 kg CO2e (68.61% of total emissions). On-site rooftop solar photovoltaic generation under INIT_SOLAR_PH1 supplied an additional 312,400 kWh of zero-carbon power, preventing 256,168 kg CO2e of additional grid emissions. Total campus electrical consumption reached 2,362,400 kWh, establishing a 13.22% clean electricity penetration rate.

2.3 Scope 3 (Indirect Value Chain and Municipal Emissions):

Scope 3 emissions totaled 719,688 kg CO2e (29.38% of total emissions):

- Commuter Transportation: Commuting across 3,000 students and 200 staff generated 580,000 kg CO2e. Modal analysis indicates that private passenger cars generated 165,300 kg CO2e (EF_TRANS_CAR = 0.175 kg CO2e/pkm), two-wheelers generated 111,940 kg CO2e (EF_TRANS_BIKE = 0.042 kg CO2e/pkm), municipal buses generated 152,000 kg CO2e (EF_TRANS_BUS = 0.054 kg CO2e/pkm), and suburban electric trains produced 150,760 kg CO2e (EF_TRANS_TRAIN = 0.028 kg CO2e/pkm).

- Solid Waste Disposal: Municipal solid waste delivered to Deonar landfill contributed 115,000 kg CO2e based on factor EF_WASTE_LANDFILL (1.25 kg CO2e/kg waste).

- Embodied Water Treatment: Municipal pumping and wastewater processing for 48,000,000 litres accounted for 24,688 kg CO2e.

3. BUILDING-SPECIFIC ENERGY AND EMISSIONS BENCHMARKING

- Computer Laboratories (BLD_LAB, 5,200 m2): Registered the highest electricity intensity at 112 kWh/m2/year (annual total 582,400 kWh), driven by 16 computer labs, 2 primary server rooms, and continuous air conditioning.

- Academic Block A (BLD_ACA, 7,500 m2): Consumed 382,500 kWh (51.0 kWh/m2/year). Rooftop solar generated 113,600 kWh, supplying 29.7% of building demand.

- Academic Block B (BLD_ACB, 6,800 m2): Consumed 333,200 kWh (49.0 kWh/m2/year). Rooftop solar supplied 99,400 kWh (29.8% solar fraction).

- Administration Block (BLD_ADM, 5,000 m2): Consumed 228,000 kWh (45.6 kWh/m2/year), offset by 56,800 kWh of solar generation.

- Library (BLD_LIB, 3,100 m2): Consumed 168,200 kWh (54.2 kWh/m2/year), with 42,600 kWh provided by rooftop solar PV.

- Auditorium (BLD_AUD, 2,800 m2): Consumed 134,200 kWh (47.9 kWh/m2/year), characterized by sharp intermittent event cooling peaks.

- Canteen (BLD_CAN, 2,100 m2): Consumed 192,000 kWh (91.4 kWh/m2/year) alongside substantial thermal and water usage.

4. TARGET TRACKING AND STRATEGIC RECOMMENDATIONS

The audit benchmarks progress against institutional targets: TRG_NET_ZERO requires linear annual abatement of 153,125 kg CO2e; TRG_RENEWABLE mandates expanding clean energy from 14.8% to 50.0% by 2030; TRG_ZERO_WASTE demands diverting 85% of waste from landfills by 2028; and TRG_WATER_NEUTRAL requires a 30% reduction in municipal water intake by 2029. Immediate auditor recommendations mandate executing Phase-2 Solar (INIT_SOLAR_PH2) on BLD_AUD and parking canopies, deploying variable refrigerant flow chillers in BLD_LAB, expanding the electric shuttle (INIT_ELECTRIC_SHUTTLE), and scaling Canteen composting (INIT_BIOGAS_CANTEEN) to support SDG 7, SDG 11, SDG 12, and SDG 13.

INSTITUTIONAL AUDIT & COMPLIANCE VERIFICATION

*This statutory record is indexed in the EcoClock AI Knowledge Graph repository under Canonical ID **DOC_AUD_002**. All empirical utility telemetry, carbon accounting estimates, and SDG alignments articulated in this document have been reviewed and verified by the VESIT Campus Sustainability Taskforce, Department of Computer Engineering (DEP_CMPN), Department of Information Technology (DEP_INFT), Department of Electronics & Telecommunication (DEP_EXTC), Department of Artificial Intelligence & Data Science (DEP_AIDS), and Campus Facilities (DEP_ADMN). Published by: Vivekananda Education Society's Institute of Technology (VESIT Green Campus), Hashu Advani Memorial Complex, Collector's Colony, Chembur, Mumbai 400074.*